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ELECTRONIC BRAKE APPARATUS

Abstract

An electronic brake apparatus is disclosed. The electronic brake apparatus includes: an input unit configured to move in a first direction according to a movement of a brake pedal; an output unit configured to be pressed according to a movement of the input unit and press a piston of a master cylinder connected to the output unit in the first direction; an electric booster including a motor, a rack configured to press the output unit, and a motion conversion unit configured to convert a rotational motion of the motor into a linear motion of the rack and move the rack forward or backward in the first direction; and an electronic control unit configured to control an operation of the motor according to a movement of the input unit, wherein the electronic control unit is positioned to one side in second direction of the electric booster.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to an electronic brake apparatus, and more particularly, to an electronic brake apparatus capable of simplifying an electronic control structure of an electric booster for adjusting pressure of a master cylinder.

BACKGROUND ART

[0002] Recently, hybrid vehicles, fuel cell vehicles, and electric vehicles are being actively developed to improve fuel efficiency and reduce exhaust gas emissions. These vehicles are necessarily equipped with a brake system, i.e., a brake apparatus. Herein, the brake apparatus of the brake system refers to an apparatus that functions to reduce the speed of the running vehicle or stop the vehicle.

[0003] Because the brake system should supply the suction pressure of the vehicle engine to a vacuum booster in order to create a vacuum, the brake system needs to be equipped with a separate engine or a vacuum generator.

[0004] However, Electronic Vehicles (EV) that do not have engines which are common vacuum generating sources or hybrid vehicles in which the engines do not always operate have a problem in that it is difficult to generate back pressure that satisfies the braking force of the vehicles if separate vacuum generators are not installed.

[0005] Therefore, recently, a brake system including an electric booster that uses a motor instead of a vacuum booster is being developed.

[0006] In an electronic brake apparatus including an electric booster, the pedal effort sensor detects, when a driver steps on the brake pedal, the driver's pedal effort and the Electronic Control Unit (ECU) drives the motor of the electric booster based on a detection value of the pedal effort sensor to transfer a boosted force to the master cylinder. The master cylinder transmits a hydraulic force toward the wheel brakes based on the boosted force transferred from the electric booster.

[0007] For this operation, the electronic control unit of the electronic brake apparatus needs to be connected to the pedal effort sensor for detecting a pedal effort or a pedal travel sensor for detecting a movement of the pedal by a pedal effort, and a motor sensor for detecting a rotation of the motor. To this end, it is necessary to position the sensors at the respective locations of the brake apparatus and electrically connect the sensors to the control circuit of the electronic control unit. In the case in which the sensors are far away from the control circuit of the electronic control unit, wirings for electrically connecting them are provided and connectors for coupling the wirings with each other are separately provided, which makes the structure complex and results in low performance in assembly.

DISCLOSURE

Technical Problem

[0008] The disclosure provides an electronic brake system capable of simplifying an electronic control structure of an electric booster that adjusts pressure of a master cylinder.

Technical Solution

[0009] An electronic brake apparatus according to an aspect of the disclosure may include: an input unit configured to move in a first direction according to a movement of a brake pedal; an output unit configured to be pressed according to a movement of the input unit and press a piston of a master cylinder connected to the output unit in the first direction; an electric booster including a

motor, a rack configured to press the output unit, and a motion conversion unit configured to convert a rotational motion of the motor into a linear motion of the rack and move the rack forward or backward in the first direction; and an electronic control unit configured to control an operation of the motor according to a movement of the input unit, wherein the electronic control unit is positioned to one side in second direction of the electric booster.

[0010] The electronic control unit may include: a motor position sensor configured to detect a rotation of the motor; a pedal travel sensor configured to detect a movement of the input unit; and a Printed Circuit Board (PCB) configuring an electronic circuit for controlling the motor.

[0011] A rotation shaft of the motor may be aligned in a second direction, the PCB may be positioned to one side in second direction of the motor, and the motor position sensor may be positioned to one side in second direction of the motor on the PCB to detect a rotation of the motor.

[0012] The electronic brake apparatus may further include: a sensor rod connected to the input unit, extending in the second direction, and moving together with the input unit; and a PTS magnet provided at one end of the sensor rod, wherein the pedal travel sensor may be provided to one side in second direction of the PTS magnet on the PCB to detect a movement of the PTS magnet.

[0013] The rack may include a hollow portion formed in the first direction, the input unit may be positioned inside the hollow portion of the rack and move in the first direction, and the sensor rod may be connected to the input unit between the output unit and the rack.

[0014] The PCB may include a first PCB and a second PCB, the motor position sensor may be positioned on the first PCB, the pedal travel sensor may be positioned on the second PCB, and the first PCB may be electrically connected to the second PCB.

[0015] The output unit may include: an output rod configured to press the piston of the master cylinder; a reaction disk configured to be pressed by the input unit and press the output rod; and a boost body configured to be pressed by the rack or the input unit and press the reaction disk.

[0016] When the rack moves forward in the first direction to press the boost body, the rack may come into contact with the input unit and move the input unit forward in the first direction.

[0017] When the brake pedal is pressed to move in the first direction, the input unit may move independently from the rack.

[0018] While the brake pedal is not pressed, the input unit may be spaced a preset gap from the reaction disk.

Advantageous Effects

[0019] An electronic brake apparatus according to an embodiment may reduce a package size and simplify a connection structure for sensors by positioning an integrated structure of a motor position sensor and a pedal travel sensor to one side of an electric booster on a Printed Circuit Board (PCB) that configures an electronic circuit for control.

Description

DESCRIPTION OF DRAWINGS

[0020] FIG. 1 is a perspective view schematically showing an electronic brake apparatus according to an embodiment of the disclosure.

[0021] FIG. 2 is a perspective view schematically showing an internal configuration of an electronic brake apparatus according to an embodiment of the disclosure.

[0022] FIG. 3 is a front view schematically showing an internal configuration of an electronic brake apparatus according to an embodiment of the disclosure.

[0023] FIG. 4 is a side view schematically showing an internal configuration of an electronic brake apparatus according to an embodiment of the disclosure.

[0024] FIG. 5 is a cross-sectional view showing the internal configuration of the electronic brake apparatus shown in FIG. 3, taken along line A-A'.

[0025] FIGS. **6** to **9** are operation diagrams schematically showing operations of an electronic brake apparatus according to an embodiment of the disclosure.

[0026] FIG. **10** is a partially cross-sectional view schematically showing a coupled state of an input unit, an output unit, and a rack of an electronic brake apparatus according to an embodiment of the disclosure.

MODES OF THE INVENTION

[0027] Throughout this specification, like reference numerals will refer to like components. The present specification does not describe all elements of embodiments, and descriptions about content being general in the technical art to which the disclosure belongs or overlapping content between the embodiments will be omitted. As used herein, the terms “portion”, “part”, “module”, “member” or “block” may be implemented as software or hardware, and according to embodiments, a plurality of “portions”, “parts”, “modules”, “members” or “blocks” may be implemented as a single component, or a single “portion”, “part”, “module”, “member” or “block” may include a plurality of components.

[0028] Through this specification, it will be understood that when a certain part is referred to as being “connected” to another part, it can be directly or indirectly connected to the other part. When a part is indirectly connected to another part, it may be connected to the other part through a wireless communication network.

[0029] Also, it will be understood that when a certain part “includes” a certain component, the part does not exclude another component but can further include another component, unless the context clearly dictates otherwise.

[0030] The terms “first”, “second”, etc., are used only to distinguish one component from another, and components should not be limited by these terms.

[0031] The singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

[0032] Hereinafter, an operation principle and embodiments of the disclosure will be described with reference to the accompanying drawings.

[0033] FIG. **1** is a perspective view schematically showing an electronic brake apparatus according to an embodiment of the disclosure.

[0034] Referring to FIG. **1**, an electronic brake apparatus **1** according to an embodiment of the disclosure may include: a master cylinder **20** in which a piston **21** presses brake oil to generate hydraulic pressure; a reservoir **30** that supplies brake fluid to the master cylinder **20**; and a booster **10** that drives a motor in response to a driver's brake pedal operation to press the piston **21** of the master cylinder **20**. The booster **10** may be used to transfer a boosted force to the master cylinder **20** in response to a driver's pedal stroke by using a motor, and may include: an input unit **100** that moves in a first direction (X direction) according to a movement of a brake pedal; an output unit **200** that is pressed according to a movement of the input unit **100** to press the piston **21** of the master cylinder **20** connected to the output unit **200** in the first direction; an electric booster **300** including a motor **310**, a rack **320** for pressing the output unit **200**, and a motion conversion unit **330** for converting a rotational motion of the motor **310** into a linear motion of the rack **320** to move the rack **320** forward or backward in the first direction; and an electronic control unit **400** that controls an operation of the motor **310** according to a movement of the input unit **100**.

[0035] FIG. **2** is a perspective view schematically showing an internal configuration of an electronic brake apparatus according to an embodiment of the disclosure, FIG. **3** is a front view schematically showing an internal configuration of an electronic brake apparatus according to an embodiment of the disclosure, FIG. **4** is a side view schematically showing an internal configuration of an electronic brake apparatus according to an embodiment of the disclosure, and FIG. **5** is a cross-sectional view showing the internal configuration of the electronic brake apparatus shown in FIG. **3**, taken along line A-A'.

[0036] In FIGS. **2** to **5**, structures and positions of the input unit **100**, the output unit **200**, the

electric booster **300**, and the electronic control unit **400** of the electronic brake apparatus shown in FIG. **1** are shown.

[0037] The electronic brake apparatus **1** may include: the input unit **100** that moves in the first direction (X direction) according to a movement of the brake pedal; the output unit **200** that is pressed according to a movement of the input unit **100** to press the piston **21** of the master cylinder **20** connected to the output unit **200** in the first direction; the electric booster **300** including the motor **310**, the rack **320** for pressing the output unit **200**, and the motion conversion unit **330** for converting a rotational force of the motor **310** into a linear motion of the rack **320** to move the rack **320** forward or backward in the first direction; and the electronic control unit **400** that controls an operation of the motor **310** according to a movement of the input unit **100**.

[0038] In this case, the electronic control unit **400** may include: a motor position sensor **410** that detects a rotation of the motor **310**; a pedal travel sensor **420** that detects a movement of the input unit **100**; and a Printed Circuit Board PCB **430** that configures an electronic circuit for controlling the motor **310**.

[0039] The master cylinder **20** may transfer a hydraulic force to a wheel brake (not shown) based on a boosted force transferred from the booster **10**. At this time, brake oil stored in the reservoir **30** may be used. The hydraulic force transferred from the master cylinder **20** may mean pressure of the brake oil generated from the master cylinder **20** toward the wheel brake in order to generate a desired friction braking force at each wheel brake.

[0040] The master cylinder **20** may include the first piston **21**, a second piston, a first spring, and a second spring to have two hydraulic circuits.

[0041] The master cylinder **20** may have the two hydraulic circuits to secure safety when a failure occurs. For example, a first circuit of the two hydraulic circuits may be connected to a right front wheel and a left rear wheel of a vehicle, and the other circuit may be connected to a left front wheel and a right rear wheel. Alternatively, unlike this, it is general that the first circuit of the two hydraulic circuits is connected to the two front wheels and the other circuit is connected to the two rear wheels. As such, configuring two circuits independently may be aimed to brake the vehicle even when any circuit fails.

[0042] The first piston **21** of the master cylinder **20** may compress the first spring, and the second piston may compress the second spring sequentially. Accordingly, the brake oil inside the master cylinder **20** may be compressed to generate brake hydraulic pressure.

[0043] The input unit **100** may include: a push rod **110** connected to the brake pedal and moving in the first direction according to a movement of the brake pedal; and a control plunger **120** moving according to a movement of the push rod **110** to press a reaction disk **220** which will be described below. The push rod **110** may include a clevis to be connected to the brake pedal, as shown.

[0044] The output unit **200** may include: an output rod **210** that presses the piston **21** of the master cylinder **20**; the reaction disk **220** that is pressed by the input unit **100** to press the output rod **210**; a boost body **230** that is pressed by the rack **320** or the input unit **100** to press the reaction disk **220**; an output rod retainer **240** that supports the output rod **210**; and an adjuster **250** that adjusts a gap between the output rod **210** and the piston **21** of the master cylinder **20**. The output unit **200** may move forward according to a forward movement of the rack **320** to move the piston **21** of the master cylinder **20** forward, thereby increasing pressure of the master cylinder **20**, and the output unit **200** may move backward according to a backward movement of the rack **320** to move the piston **21** of the master cylinder **20** backward, thereby decreasing pressure of the master cylinder **20**. Or, the output unit **200** may be in contact with the input unit **100** and move forward according to a forward movement of the input unit **100** by a pedal stroke to move the piston **21** of the master cylinder **20** forward, thereby increasing pressure of the master cylinder **20**, and the output unit **200** may move backward according to a backward movement of the input unit **100** to move the piston **21** of the master cylinder **20** backward, thereby decreasing pressure of the master cylinder **20**.

[0045] Meanwhile, a return spring **700** may be connected to the output rod retainer **240**. The return

spring **700** may be provided between the output rod retainer **240** coupled to the output rod **210** and the master cylinder **20** to move the output rod **210** backward while the output rod **210** is not pressed by the input unit **100** or the rack **320**, thereby decreasing pressure of the master cylinder **20**.

[0046] The motor **310** may be a motor capable of selecting a forward rotation which is a rotation in one direction and a backward rotation which is a rotation in the opposite direction. In the drawings, for convenience of description, a stator, a rotor, and a driving circuit for driving the motor **310** are omitted, and only a rotating shaft of the motor **310** is shown. The rack **320** may be movable linearly in such a way as to move forward or backward in the first direction. The motion conversion unit **330** may convert a rotation motion of the motor **310** into a linear motion of the rack **320**. The motion conversion unit **330** may be a pinion gear of which one side is connected to the rotating shaft of the motor **310** and another side is connected to the rack **320**.

[0047] The rack **320** may have a hollow portion formed in the first direction. In this case, the input unit **100** may be positioned inside the hollow portion of the rack **320** and move in the first direction. As such, because the input unit **100** that presses the output rod **210** is positioned inside the hollow portion provided in the rack **320**, a size of a pressing structure may be reduced.

[0048] The motion conversion unit **330** may convert a rotation motion of the motor **310** into a linear motion of the rack **320**, while amplifying power generated from the motor **310** through a reduction device and transferring the power to the rack **320**. For example, the motion conversion unit **330** may include a worm shaft **331** connected to the rotating shaft to rotate according to driving of the motor **310** and having a worm gear formed on the outer circumferential surface, a worm wheel **332** having gear teeth engaged with the worm gear of the worm shaft **331**, and a pinion gear **333** which rotates by a rotation of the worm wheel **332** and is engaged with the rack **320**. That is, the worm wheel **332** may rotate together with the pinion gear **333** by a rotation force transferred from the worm shaft **331** by a rotation of the motor **310**, and the rack **320** engaged with the pinion gear **333** may move forward or backward in the first direction to press or release the output rod **210**.

[0049] Meanwhile, the electric booster **300** may further include a bush **340** that supports the rack **320** and prevents departure of the pinion gear **33**.

[0050] The electronic control unit (ECU) **400** may detect, when a driver steps on the brake pedal, a pedal stroke through the pedal travel sensor **420**, rotate the motor **310** in one direction to generate target brake pressure corresponding to the detected pedal stroke, and thus move the rack **320** forward through the motion conversion unit **330**. Accordingly, the rack **320** may move the output unit **200** forward and the output unit **200** moved forward may press the piston **21** being in close contact with the output unit **200**. Therefore, the brake oil inside the master cylinder **20** may be compressed and thus pressure of the master cylinder **20** may increase.

[0051] Also, the electronic control unit **400** may rotate, when the driver takes his/her foot off the brake pedal to release the brake pedal, the motor **310** in the opposite direction to move the rack **320** backward through the motion conversion unit **330**. Accordingly, the output unit **200** may move backward according to the backward movement of the rack **320** and thus pressure of the master cylinder **20** may decrease.

[0052] Meanwhile, referring to FIGS. 2 to 5, the electronic brake apparatus **1** according to an embodiment of the disclosure may further include a sensor rod **500** connected to the input unit **100**, extending in a second direction (Y direction), and moving together with the input unit **100**; and a PTS magnet **600** provided at one end of the sensor rod **500**. That is, the PTS magnet **600** positioned in the second direction from the input unit **100** may be connected to the input unit **100** via the sensor rod **500**, and move in the first direction together with the input unit **100** moving in the first direction.

[0053] The pedal travel sensor **420** may be provided to one side in second direction of the PTS magnet **600**, on the PCB **430**, and detect a movement of the PTS magnet **600** to detect a movement

of the input unit **100**.

[0054] Meanwhile, the motor position sensor **410** may detect a position of the motor **310**, that is, at least one of an absolute angle and a relative angle according to a rotation of the motor **310**, and provide the detected result to the electronic control unit **400**. To this end, the motor position sensor **410** may include at least one of a relative angle sensor that detects a relative rotation angle of the motor **310** and an absolute angle sensor that detects an absolute rotation angle of the motor **310**. To this end, a MPS magnet **315** may be provided in the rotating shaft or the rotor of the motor **310** to rotate according to a rotation of the motor **310**, and the motor position sensor **410** may detect a rotation of the motor **310** by detecting a rotation of the MPS magnet **315**.

[0055] In this case, as shown in FIGS. **2** to **5**, the rotating shaft of the motor **310** may be aligned in the second direction (Y direction) that is perpendicular to the first direction (X direction) as a movement direction of the rack **320**, the PCB **430** may be positioned to one side in second direction of the motor **310**, and the motor position sensor **410** may be provided to one side in second direction of the motor **310**, on the PCB **430**.

[0056] That is, as shown in FIGS. **2** to **5**, because the motor position sensor **410** is positioned to one side in second direction of the motor **310**, the pedal travel sensor **420** is also positioned around the PTS magnet **600** located in the second direction from the input unit **100**, and the motor position sensor **410** and the pedal travel sensor **420** are located on the PCB configuring an electronic circuit, the entire electronic control unit **400** may be positioned to one side in second direction of the electric booster **300**.

[0057] As such, by positioning an integrated structure of the motor position sensor **410** and the pedal travel sensor **420** to one side of the electric booster **300** on the PCB **430** configuring the electronic circuit for control, a package size may be reduced and a connection structure for sensors may be simplified.

[0058] Meanwhile, the PCB **430** may include a first PCB **430a** and a second PCB **430b**, the motor position sensor **410** may be positioned on the first PCB **430a**, the pedal travel sensor **420** may be positioned on the second PCB **430b**, and the first PCB **430a** may be electrically connected to the second PCB **430b**.

[0059] As described above, although the PTS magnet **600** is located close to the PCB **430** positioned to one side in second direction of the electric booster **300** by the sensor rod **500** extending in the second direction from the input unit **100**, sensitivity of the pedal travel sensor **420** may be lowered by a movement of the PTS magnet **600** due to deformation and vibration of the sensor rod **500** as a length of the sensor rod **500** increases. Accordingly, in an embodiment of the disclosure, the electronic control unit **400** may be configured by positioning the motor position sensor **410** and the pedal travel sensor **420** respectively on two PCBs **430a** and **430b** and electrically connecting the two PCBs **430a** and **430b** to each other. As such, by configuring the electronic control unit **400** through the two PCBs **430a** and **430b**, the length of the sensor rod **500** may be reduced and the electronic control unit **400** including the motor position sensor **410** and the pedal travel sensor **420** may be positioned to one side of the electric booster **300**.

[0060] Meanwhile, a magnet spring **800** may be provided between the PTS magnet **600** and a case of the booster **10**. The magnet spring **800** may be pressed when the PTS magnet **600** moves forward in the first direction to minimize a gap of the PTS magnet **600** and thus increase accuracy of the pedal travel sensor **420**.

[0061] Hereinafter, an operation of the electronic brake apparatus **1** according to the disclosure will be described with reference to FIGS. **6** to **10**.

[0062] FIGS. **6** to **9** are operation diagrams schematically showing operations of an electronic brake apparatus according to an embodiment of the disclosure.

[0063] FIG. **6** shows a state when no pedal effort is applied to the brake pedal of the electronic brake apparatus **1**.

[0064] When no pedal effort is applied to the pedal, the input unit **100**, the rack **320**, and the output

unit **200** may be positioned not to press the piston **21** of the master cylinder **20**.

[0065] FIG. **7** shows a state in which the input unit **100** and the rack **320** press the output unit **200** when a pedal effort is applied to the brake pedal of the electronic brake apparatus **1**.

[0066] When a driver applies a pedal effort to the brake pedal, the input unit **100** may move forward and the PTS magnet **600** connected through the sensor rod **500** may also move forward. The forward movement of the PTS magnet **600** may be detected by the pedal travel sensor **420**, and the electronic control unit **400** may drive the motor **310** to move the rack **320** forward. As such, because the rack **320** moves forward by driving of the motor **310** and the rack **320** presses the output unit **200**, the output unit **200** may be strongly pressed with a small force applied to the brake pedal by the driver to press the input unit **100**, thereby increasing hydraulic pressure generated in the master cylinder **20**. At this time, the control plunger **120** of the input unit **100** may come into contact with the reaction disk **220** of the output unit **200** and the driver may receive pedal feeling through the brake pedal.

[0067] FIG. **8** shows a state in which the rack **320** presses the output unit **200** according to Adaptive Cruise Control (ACC) braking by the electronic control unit **400** of the electronic brake apparatus **1**.

[0068] The electronic brake apparatus **1** according to the disclosure may perform braking according to determination by the electronic control unit **400** even when a driver applies no pedal effort to the brake pedal. During control such as ACC to maintain a constant speed of the vehicle, braking may be, when the speed of the vehicle exceeds preset speed, performed to reduce the speed of the vehicle. At this time, the electronic control unit **400** may drive the motor **310** to move the rack **320** forward. The rack **320** moved forward by the driving of the motor **310** may press the output unit **200**, thereby performing braking without a pedal effort applied by the driver.

[0069] Meanwhile, when the rack **320** moves in the first direction to press a boost body **230** of the output unit **200**, the rack **320** may come into contact with the input unit **100** and move the input unit **100** in the first direction.

[0070] For example, as shown in FIG. **8**, the sensor rod **500** may be connected to the control plunger **120** of the input unit **100** between the output unit **200** and the rack **320**. When the rack **320** moves forward by driving of the motor **310**, the rack **320** may press the boost body **230** in the first direction, and simultaneously, one end of the rack **320** may come into contact with the sensor rod **500** to move the control plunger **120** connected to the sensor rod **500** forward in the first direction. Accordingly, the push rod **110** and the brake pedal connected to the push rod may also move forward together in the first direction by the forward movement of the control plunger **120**. In the case in which the brake pedal remains without moving forward together while the rack **320** is driven by the motor **310** without a pedal stroke by a driver, no reaction force may be applied to the pedal upon a sudden pedal stroke by the driver, which may result in sudden braking. Accordingly, to prevent this, it may be preferable that, while the rack **320** moves forward, the input unit **100** also moves forward together.

[0071] FIG. **9** shows a fallback driving state of the electronic brake apparatus **1**.

[0072] In the electronic brake apparatus **1** according to an embodiment of the disclosure, the rack **320** may not operate due to various causes. The rack **320** may not operate when the electronic control unit **400** malfunctions, when the motor **310** does not operate, when no power is transmitted through the motion conversion unit **330**, or when the rack **320** itself malfunctions. Even while the rack **320** does not operate, the driver should be able to perform braking through the brake pedal when he/she needs to brake the vehicle.

[0073] In the disclosure, while the rack **320** does not operate, the input unit **100** may directly press the output unit **200** to press the piston **21** of the master cylinder **20**, as shown in FIG. **9**. Thereby, the master cylinder **20** may generate hydraulic pressure to drive the wheel brake, thereby performing braking.

[0074] At this time, when the brake pedal is pressed to move in the first direction, the input unit

100 may move independently from the rack **320**. Because the input unit **100** is movable independently from the rack **320** even while the rack **320** does not operate, the input unit **100** may directly press the output unit **200**, thereby performing braking.

[0075] As described above, the sensor rod **500** may be connected to the control plunger **120** of the input unit **100** between the output unit **200** and the rack **320**. Accordingly, when the rack **320** moves forward in the first direction, one end of the rack **320** may come into contact with the sensor rod **500** to move the control plunger **120** connected to the sensor rod **500** forward in the first direction, whereas, when the control plunger **120** moves forward in the first direction, the sensor rod **500** spaced from the one end of the rack **320** may move forward independently.

[0076] FIG. **10** is a partially cross-sectional view schematically showing a coupled state of an input unit, an output unit, and a rack of an electronic brake apparatus according to an embodiment of the disclosure.

[0077] As shown in FIG. **10**, the reaction disk **110** may be provided in the output unit **200**. The reaction disk **220** may be pressed by the control plunger **120** of the input unit **100** and the boost body **230**, and when the reaction disk **220** is pressed by the control plunger **120** or the boost body **230**, the reaction disk **220** may function to transfer a pressing force to the output rod **210**.

[0078] While the brake pedal is not pressed, the control plunger **120** of the input unit **100** may be spaced a preset gap **d1** (Lap gap) from the reaction disk **220**. As such, because the gap **d1** is provided between the control plunger **120** and the reaction disk **220**, Jump in in which, at an initial stage at which the pedal is pressed, no output is generated even though a pedal effort is applied and when the control plunger **120** comes into contact with the reaction disk **220**, an output increases instantaneously may occur.

[0079] Meanwhile, while the brake pedal is not pressed, the control plunger **120** of the input unit **100** and/or the sensor rod **500** may be spaced a preset gap **d2** (Boost gap) from the boost body **230**. During a normal operation, because, when the control plunger **120** moves forward in the first direction, the rack **320** moves forward by driving of the motor **310** to press the boost body **230** and move the boost body **230** forward and the boost body **230** presses the reaction disk **220**, the control plunger **230** and/or the sensor rod **500** may be spaced from the boost body **230**. However, when the rack **320** continues to move forward and reaches a driving limit, the rack **320** may no longer move forward. At this time, when a driver continues to press the brake pedal, the input unit **100** may further move forward and the control plunger **120** and/or the sensor rod **500** may directly come into contact with the boost body **230** and press the boost body **230**. As such, when the input unit **100** directly presses the boost body **230**, a driving force by the rack **320** may be no longer added. Accordingly, an output of the output unit **200** amplified through the booster **10** until the rack **320** reaches the driving limit may become equal to a magnitude of a pedal effort applied to the input device **100**, and Knee-point may appear in a graph showing relationship between a pedal effort of the input unit **100** and an output of the output unit **200**.

[0080] Meanwhile, even in fallback driving, because the rack **320** does not move forward, the input unit **100** may directly press the boost body **230** and the boost body **230** may press the reaction disk **220** to move the output rod **210** forward.

[0081] Through the above-described structure, the electronic brake apparatus according to an embodiment of the disclosure may reduce a package size and simplify a connection structure for sensors by positioning an integrated structure of the motor position sensor and the pedal travel sensor to one side of the electric booster on the PCB that configures the electronic circuit for control.

[0082] So far, the disclosed embodiments have been described with reference to the accompanying drawings. It will be understood by one of ordinary skill in the technical art to which the disclosure belongs that the disclosure can be embodied in different forms from the disclosed embodiments without changing the technical spirit and essential features of the disclosure. Thus, it should be

understood that the disclosed embodiments are merely for illustrative purposes and not for limitation purposes.

Claims

1. An electronic brake apparatus comprising: an input unit configured to move in a first direction according to a movement of a brake pedal; an output unit configured to be pressed according to a movement of the input unit and press a piston of a master cylinder connected to the output unit in the first direction; an electric booster including a motor, a rack configured to press the output unit, and a motion conversion unit configured to convert a rotational motion of the motor into a linear motion of the rack and move the rack forward or backward in the first direction; and an electronic control unit configured to control an operation of the motor according to a movement of the input unit, wherein the electronic control unit is positioned to one side in second direction of the electric booster.
 2. The electronic brake apparatus of claim 1, wherein the electronic control unit comprises: a motor position sensor configured to detect a rotation of the motor; a pedal travel sensor configured to detect a movement of the input unit; and a Printed Circuit Board (PCB) configuring an electronic circuit for controlling the motor.
 3. The electronic brake apparatus of claim 2, wherein a rotation shaft of the motor is aligned in a second direction, the PCB is positioned to one side in second direction of the motor, and the motor position sensor is positioned to one side in second direction of the motor on the PCB and configured to detect a rotation of the motor.
 4. The electronic brake apparatus of claim 2, further comprising: a sensor rod connected to the input unit, extending in the second direction, and moving together with the input unit; and a PTS magnet provided at one end of the sensor rod, wherein the pedal travel sensor is provided to one side in second direction of the PTS magnet on the PCB and configured to detect a movement of the PTS magnet.
 5. The electronic brake apparatus of claim 4, wherein the rack includes a hollow portion formed in the first direction, the input unit is positioned inside the hollow portion of the rack and moves in the first direction, and the sensor rod is connected to the input unit between the output unit and the rack.
 6. The electronic brake apparatus of claim 4, wherein the PCB includes a first PCB and a second PCB, the motor position sensor is positioned on the first PCB, the pedal travel sensor is positioned on the second PCB, and the first PCB is electrically connected to the second PCB.
 7. The electronic brake apparatus of claim 1, wherein the output unit includes: an output rod configured to press the piston of the master cylinder; a reaction disk configured to be pressed by the input unit and press the output rod; and a boost body configured to be pressed by the rack or the input unit and press the reaction disk.
 8. The electronic brake apparatus of claim 7, wherein, when the rack moves forward in the first direction to press the boost body, the rack comes into contact with the input unit and moves the input unit forward in the first direction.
 9. The electronic brake apparatus of claim 8, wherein, when the brake pedal is pressed to move in the first direction, the input unit moves independently from the rack.
 10. The electronic brake apparatus of claim 7, wherein, while the brake pedal is not pressed, the input unit is spaced a preset gap from the reaction disk.
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